Pyung Kim

Postdoctoral Scholar

Center for Spatial Studies and Data Science
University of California, Santa Barbara
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ACADEMIC APPOINTMENTS

Postdoctoral Scholar 2026 – Present
Center for Spatial Studies and Data Science, University of California, Santa Barbara

PROFESSIONAL AFFILIATIONS

Research Affiliate 2024 – 2025
Spatial Health AI Research Partnership (SHARP), The University of Texas at Dallas

Instructor 2023 – 2025
Korea Social Science Data Archive (KOSSDA), Seoul National University

Research Assistant 2020
Korea Institute of Public Administration (KIPA)

EDUCATION

Ph.D. in Public Policy and Political Economy 2025
The University of Texas at Dallas (Brian J.L. Berry Scholarship, Highest Honor)
Dissertation: *"Long-Term Effects of Neighborhoods on Educational Attainment by Disability Status"*

M.S. in Geospatial Information Sciences 2025
The University of Texas at Dallas
Master's Research: *"Estimating Indoor Radon Risk: A Kriging Approach with Geological Predictors"*

M.A. in Public Administration 2020
Yonsei University (Brain Korea 21 PLUS Scholarship, National Research Foundation)

B.A. in Public Administration 2018
Ajou University (National Excellence Scholarship, Deputy Prime Minister of Korea)
Julius-Maximilian University of Würzburg, Germany (Exchange Student, 2016 – 2017)

RESEARCH AREAS

Substantive	Health Policy, Social Policy, Climate Policy, Neighborhood Effects, Disability
Methodological	GIS, Geo AI, Spatial Data Science, Machine Learning, Causal Inference

PEER-REVIEWED PUBLICATIONS *[*] Corresponding author, [†] Student co-author*

6. **Kim, P.**, Choi, S.[†], Kim, D., & Lee, C[*]. (2026). Bayesian spatio-temporal modeling for policy evaluation: Sensitivity of policy effect estimates in the context of COVID-19 stay-at-home orders. *PLoS One, 21(2), e0339196.*

5. **Kim, P.** [*]**,** Kim D., Scotch R., Jeong D.[†], & Kowalske K. (2025). Geospatial analysis of community-level social and environmental barriers for adult burn injury survivors in North Texas. *Burns*, 51(5), 107512.

4. Hong, S.[*], Jeong, D.[†], & **Kim, P.** (2024). Have offender demographics changed since the COVID-19 Pandemic? Evidence from money mules in South Korea**.** *Journal of Criminal Justice*, 91, 102156.

3. **Kim, P.** [*], Maxwell, S., Parijat, N.[†], Kim, D., & McNeely, C. L. (2024). Targeted tick-borne disease recognition: Assessing risk for improved public health. *Healthcare*, 12(10), 984.

2. Maxwell, S. P., Brooks, C., **Kim, P.**, Kim, D., McNeely, C. L., & Thomas, K[*]. (2023). Understanding habitats and environmental conditions of white-tailed deer population density and public health data to aid in assessing human tick-borne disease risk. *Microorganisms*, 11(4), 865.

1. **Kim, P.** & Bae, H.[*] (2022). Do firms respond differently to the carbon pricing by industrial sector? How and why? A comparison between manufacturing and electricity generation sectors using firm-level panel data in Korea. *Energy Policy*, 162, 112773.

UNDER REVIEW

"Growing Up in Disability-Dense School Districts: Long-Term Impacts of Childhood Exposure on Educational Attainment" (**sole author**)

"Are Neighborhood Effects Larger for Children with Disabilities? Evidence from Texas" (**sole author**)

"A Geography of Administrative Burden: Social Security Field Office Accessibility and the Consequences of Closures" (**first author**)

"Does Carbon Pricing Outperform Command-and-Control Regulation? Firm-Level Evidence from Korea's Dual Regulatory Framework" (**sole author**)

"mHealth Intervention Integrating Personal PM2.5 Monitoring and Deep Learning to Reduce Pediatric Asthma Exacerbations: A Pilot Study" (**first author**, preprint available at JMIR)

"Modeling Policy Diffusion Across Space and Time: A Bayesian Spatial-Temporal Approach" (**first author**)

"Spatial Analysis of Preventive Maternal and Child Oral Health Services in Kilifi County, Kenya: Implications for MCH Workforce Training" (**first author**)

WORKING PAPERS

"Medicaid Expansion and Cancer Outcomes: Heterogeneous Effects by Geographic Access to Oncology Centers" (with Kristine Hahm)

"Where Does the Evidence Come From? Mapping the Geographic Footprint of Evidence from Federal Clearinghouses" (with Sarah Bardin & Peter Kedron)

"Do Better Neighborhoods Always Improve Student Achievement? Heterogeneous Effects of Residential Mobility" (with Peter Kedron)

"Spatial Turn in Causal Inference" (with Lixi Xu, Lei Song, & Peter Kedron)

EXTERNAL GRANTS *Non-U.S. awards are reported in USD using award-date exchange rates.*

I have participated in awarded grants totaling $3,717,126, of which my recognized contribution was $32,800.

[PI / Co-PI]

1. What Is the Relationship Between Adolescent Neighborhood Environments and Long-Run Labor Market Outcomes for Children with Disabilities? 2024 – 2025
   - **PI: Pyung Kim**, UTD
   - *Social Security Administration via Retirement and Disability Research Center at UW-Madison* ($6,000; Recognition: $6,000)

[Key Personnel]

4. Interventional Study to Prevent Exacerbation of Atopic Dermatitis by Exposure to Particulate Matter 2024 – 2027
   - Role: Data Scientist
   - PI: Kangmo Ahn, *Samsung Medical Center*
   - *Korea Disease Control and Prevention Agency* ($629,630; Recognition: $4,600)

3. Geospatial Community Mapping to Assess Social and Environmental Barriers and Facilitators for Burn Survivors 2022 – 2026
   - Role: GIS Specialist
   - PI: Karen Kowalske, *UTSW Medical Center*; co-PI: Dohyeong Kim & Richard Scotch, *UTD*
   - *National Institute on Disability, Independent Living, and Rehabilitation Research* ($2,320,495; Recognition: $12,600)

2. The Capacity-Building Project for Primary Healthcare Workforce to Improve Maternal and Child Oral Health in Kilifi County, Kenya 2025 – 2026
   - Role: GIS Specialist **(field study in Kenya, October 2025)**
   - PI: Hyewon Lee, *Seoul National University*
   - *Korea International Cooperation Agency* ($440,000; Recognition: $5,600)

1. Spatial Distribution of Natural Radiation in Jeonbuk and Jeonnam Provinces 2023 – 2025
   - Role: GIS Specialist
   - PI: Yeol Rho, *Chonnam National Univ.*; co-PI: Dohyeong Kim, *UTD*
   - *National Institute of Environmental Research of Korea* ($320,000; Recognition: $4,000)

AWARDS AND HONORS

[External Awards]

Public Administration Research Conference Travel Award, *University of Central Florida* 2025

JSIT Research Competition Award, *Social Security Administration via UW-Madison* 2024

Junior Scholar Intensive Training (JSIT), *Social Security Administration via UW-Madison* 2024

President Travel Award for APPAM, *Korean Association for Policy Studies* 2023

Brain Korea 21 PLUS Scholarship, *National Research Foundation* 2018 – 2020

The Next Century Humanities Scholarship (National Excellence Scholarship), *awarded by the Deputy Prime Minister; funded by the Korea Student Aid Foundation* 2015 – 2018

[Internal Awards]

GIS Pioneer Research Scholarship, *UT Dallas & Pioneer Natural Resources* 2025

Keith Lankford Taylor Fellowship, *UT Dallas* 2023

Brian J.L. Berry Scholarship (Highest Honor), *UT Dallas* 2021 – 2025

UT Dallas Graduate Studies Scholarship, *UT Dalla* 2021 – 2025

YUPA Graduate Studies Scholarship, *Yonsei University* 2018 – 2020

President's Award for Outstanding Graduate, *Ajou University* 2018

Teaching Assistantship for Undergraduate Student, *Ajou University* 2015 – 2018

Excellence in Undergraduate Research Project, *Ajou University* 2015

Merit-based Full Tuition Scholarship, *Ajou University* 2014

Dean's list, *Ajou University* 2014; 2015

MEDIA APPEARANCES

UT Dallas News. (2026). *How Communities Shape Burn Survivors' Return to Society*. https://news.utdallas.edu/social-sciences/burn-survivors-societal-reentry-2026/

INVITED TALKS

"Long-Term Effects of Neighborhoods on Educational Attainment by Disability Status", *Geography Colloquium Lecture, University of California, Santa Barbara* May 2026 (planned)

"Using GIS to Improve Rural Health Service Planning: Lessons from Kilifi County, Kenya", *Practitioner Seminar, Kilifi Department of Health in Kilifi County, Kenya* Oct 2025

TEACHING

[Co-Instructor]

Machine Learning and Deep Learning for Social Science (with R) — Winter 2026
- *Korea Social Science Data Archive* (graduate)
- *Rating: 9.64/10*

GIS and Spatial Statistics (with QGIS, GeoDa, and R) — Summer 2025
- *Korea Social Science Data Archive* (graduate)
- *Rating: 9.91/10*

Advanced GIS and Spatial Statistics (with R) — Summer 2023
- *Korea Social Science Data Archive* (graduate)
- *Rating: 9.84/10*

Introductory GIS and Spatial Statistics (with QGIS, GeoDa, and STATA) — Summer 2023
- *Korea Social Science Data Archive* (graduate)
- *Rating: 9.54/10*

[Guest Lecture]
- Quasi-Experimental Mover Designs, Geography and Policy Lab, *UC Santa Barbara* — Feb 2026
- Analysis of Variance, undergraduate statistics course (EPPS 2302), *UT Dallas* — April 2025
- Probability Distributions, undergraduate statistics course (EPPS 2302), *UT Dallas* — Feb 2025
- Introductory Calculus, graduate statistics course (EPPS 7313), *UT Dallas* — Sep 2023

CONFERENCE PRESENTATIONS

* APPAM (Association for Public Policy Analysis and Management); APHA (American Public Health Association); ASPA (American Society for Public Administration); ACRM (American Congress of Rehabilitation Medicine); SER (Society for Epidemiologic Research); ASA (American Sociological Association)

2026 APPAM (planned)
2025 APPAM, APHA, ASA, SER
2024 APPAM, ACRM, ASPA
2023 APPAM, APHA, ASPA

RESTRICTED DATA ACCESS

2. The Impacts of School Neighborhoods on the Academic and Labor Market Outcomes of Students with Disabilities — 2024
   - *PI:* ***Pyung Kim***
   - *Student-level longitudinal data (1999 – 2023, N $\geq$ 10,000,000)*
   - *Source: Texas Schools Project*

1. Geospatial Community Mapping to Assess Social and Environmental Barriers and Facilitators for Burn Survivors — 2022
   - *PI: Karen Kowalske, UTSW Medical Center, UTD*
   - *Patient-level longitudinal data (1994 – 2022, N $\geq$ 25,000)*
   - *Source: National Institute on Disability, Independent Living, and Rehabilitation Research*

TECHNICAL SKILLS

Statistical Analysis (R, STATA)
Geospatial Analysis (R, Arc GIS, QGIS, GeoDa, Tableau)
Machine Learning / Deep Learning (R)

SERVICE

[Conference Reviewer]
APPAM Annual Fall Research Conference 2026

[Peer Reviewer]
Policy Studies Journal; Scientific Reports; Energy Policy; Energy Economics; International Journal of Finance and Economics; PLOS ONE

[University Service]
Secretary, *Korean Student Association, UT Dallas* 2023 – 2024
Vice President, *Graduate Student Assembly in Public Administration, Yonsei University* 2018 – 2019
Student Mentor, *Office of Admissions, Ajou University* 2014 – 2015